

Examination paper for TDT4175 Information Systems

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Examination date: 16 December 2019

Examination time (from-to): 9:00 – 13:00

Permitted examination support material: Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been specified. Clean copies of the chapters 1-6 used in the course also allowed. In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.

Language: English

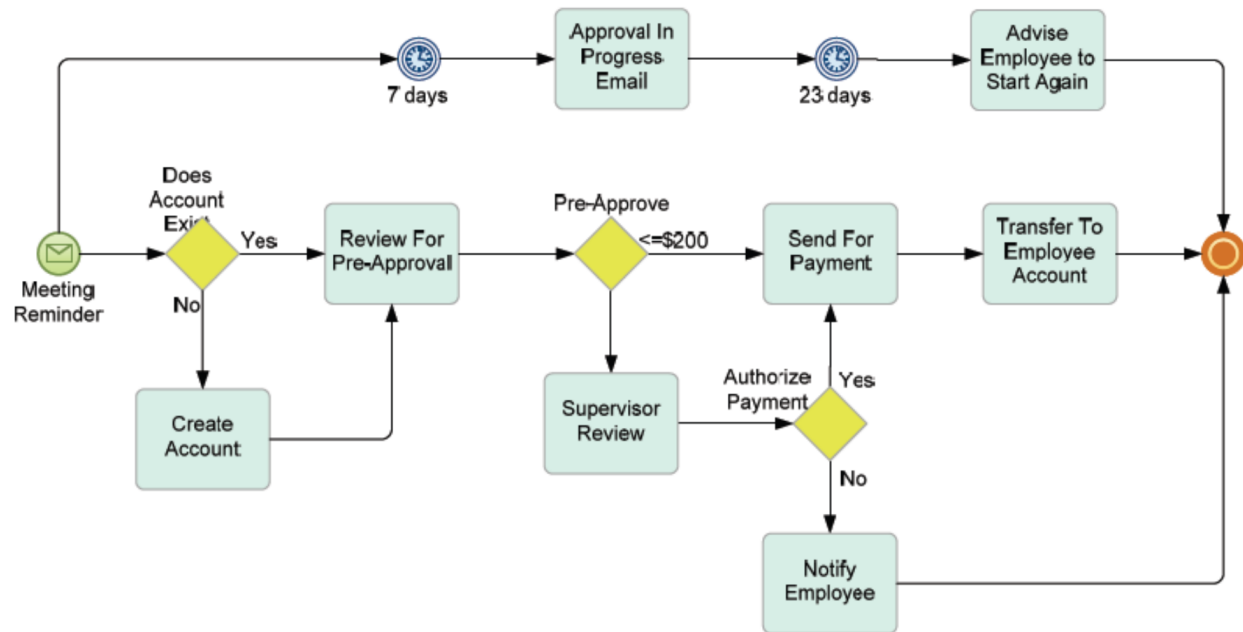
Number of pages (front pages excluded): 2

Number of pages enclosed: 3

Task 1 (20 points) - Design a sample expense reimbursement process using BPMN. This process provides for reimbursement of expenses incurred by employees for the company in which they work. Examples of reimbursements include: buying a technical book, office supplies or software etc. This process is handled by the financial department.

After the reception of a reimbursement claim, a new account must be created if the employee does not already have one. The claim is then reviewed for automatic approval. Amounts under \$200 are automatically approved, whereas amounts equal to or over \$200 require approval of the supervisor. In case of rejection, the employee must receive a rejection notice by email. The reimbursement goes to the employee's direct deposit bank account if it is automatically accepted or is accepted by the supervisor. If the request is not completed in 7 days, then the employee must receive an "approval in progress" email. If

the request is not finished within 30 days, then the process is stopped and the employee is sent an email to start the process again.



Task 2 (5 points) – List the components of a Product Lifecycle Management (PLM) system and briefly describe each of them and their functions.

Task 3 (5 points) – In Lecture 3 we discussed different types of enterprise systems. Which type of system would you use to support the processes from task 1. Motivate your choice.

Task 4 (5 points) – List the stages in Lewin's Change Model and briefly outline what each stage includes.

Task 5 (10 points) – In Lecture 2 we talked about Miles and Snow's strategy typology. List the different strategies that they distinguish **(4 of 10 points)**. In which typology would you place Google and why? Motivate your answer **(6 of 10 points)**.

Task 6 (5 points) – In Lecture 1 we talked about the dimensions of valuable information. List 3 that you believe are relevant for the case of credit card transactions and motivate your choice by briefly describing why they are important.

Task 7 (5 points) – In Lecture 3 we discussed different types of e-commerce. List 3 of these and for each give a real-life example.

Task 8 (10 points) – You have been requested to digitalize the processes of a local manufacturing company so that there is greater control of inventory, finances, customers and sales and marketing. Which type of Information System would you use to do that? Motivate your answer.

Task 9 (10 points) – You have been asked to suggest an information system for a hospital so that doctors can share their best practices and experiences with colleagues. What type of system is best suited for such a use? Describe specifically which technology you would use within this type of information systems and motivate your choice.

Task 10 (5 points) – What is the difference between re-engineering and continuous improvement?